

Chirag Singh

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[Portfolio](#) • [LinkedIn](#) • [Github](#) • [Leetcode](#)

Software Engineer with 3.5+ years of experience delivering scalable, cloud-based full stack solutions using .NET Core, React, C#, and Azure. Proven ability to architect and independently ship production-grade systems handling millions of daily events. Currently completing an MSc in Software Engineering at the University of Limerick (2026). Contributed as a mentor at the Teen-Turn programme and competed in the Lero Hack 2026 - Snapdragon: AI Unplugged Hackathon. Certified in Azure Fundamentals (AZ-900), .NET Development, and Generative AI.

Education

SEP 2025 – AUG 2026

MSc Software Engineering (Machine Learning) | University of Limerick | Limerick, Ireland

3.44 QCA (1st Semester)

AUG 2018 – NOV 2022

BTech in Information Technology | Noida Institute of Engineering & Technology | Greater Noida, India

7.98 GPA

Experience

APR 2023 – AUG 2025

Engineer | Qualitest Infotech Pvt. Ltd. | Noida, India

Client: Penn Foster Inc. | USA

1. AiGrading Application

- Developed from scratch using React, .NET (C#), and SQL, integrating with D2L Brightspace via LTI 1.3 for seamless and secure single sign-on and grade passback.
- Integrated Learnosity's Feedback Aide API to deliver AI-generated feedback on student submissions, enabling iterative resubmission workflows that supported learner mastery, used by over 75,000 learners.
- Built a workflow routing final submissions to a proprietary grading platform for manual review and quality assurance.

2. LMS Connector for D2L Brightspace Replatforming

- Developed independently an Azure serverless .NET application to process events from Desire2Learn LMS for a user base of more than 500,000, handling 1 to 2 million events daily.
- Integrated Event Hub and Service Bus for data filtering and reliable message queuing, and synchronized grades with Anthology SIS.

3. React Frontend Application with API Integration

- Built a React frontend for managing course, lesson, and program mappings, with backend database interaction via an HTTP trigger in LMS Connector, ensuring an intuitive user interface for data updates.

4. Proprietary Grading Platform

- Modified the legacy proprietary grading platform application to enhance usability and efficiency in grading student assignments for the replatformed LMS environment.

JAN 2022 – APR 2023

Graduate Engineer | Qualitest Infotech Pvt. Ltd. | Noida, India

Client: Penn Foster Inc. | USA

1. PFX 2.0 / UNOW - Legacy LMS Maintenance

- Maintained a Ruby on Rails application deployed on Heroku, ensuring system reliability and functionality.

2. D2L Print Application Optimization

- The D2L Print application serves as a widget within the Desire2Learn Learning Management System (LMS), enabling users to download course content in PDF format.
- Working independently, undertook a comprehensive redesign of the application to enhance its efficiency in generating PDFs.
- Performance optimizations were implemented to facilitate faster content downloads, significantly improving the application's responsiveness.

Academic Projects

Car Pooling System | Java, Spring Boot, React.js, PostgreSQL, Microservices | Advanced Software Architectures, UL ([click](#))

- Architected a full-stack carpooling platform using a microservices architecture (User, Ride, Payment services) with Spring Cloud Gateway, Eureka Discovery, and REST APIs, deployed on Render.
- Implemented Stripe payment integration, role-based access control, and real-time ride booking with driver/passenger workflows, serving as Project Manager & Systems Analyst on a 4-person team.

AI-Generated Image Detector | Python, PyTorch, CNN, Scikit-learn | Machine Learning, UL ([click](#))

- Built a two-phase deep learning pipeline to detect AI-generated images, progressing from a logistic regression baseline (68.5% accuracy) to a custom CNN achieving 96.18% accuracy on the CIFAKE dataset.
- Evaluated cross-dataset generalisation on Midjourney images, achieving 71.9% accuracy under domain shift, with threshold optimisation to maximise precision-recall tradeoff.

Skills

Experienced: .Net (C#), Ruby on Rails, React JS, HTML, CSS, JavaScript, Python, SQL

Familiar: Java, C/C++, Node JS

Tools: Git & GitHub, Azure (Event Hub, Service Bus, Functions), AWS, Heroku, MS Office

Activities

Mentor | Teen-Turn Programme | University of Limerick

Mentored teenage girls from underserved communities in STEM, supporting hands-on tech learning and career development

Participant | Lero Hack 2026, Snapdragon: AI Unplugged Hackathon | University of Limerick

Competed in an AI-focused hackathon, designing and building an edge AI solution under time constraints

Achievements

HighQ Award Winner - 2023 & 2024 Qualitest Infotech Pvt. Ltd. | India

Recognized for delivering high quality solutions and exceeding client expectations two years in a row.

Certifications

Introduction to Generative AI and Prompt Engineering June 2025 ([click](#))

AZ-900 Microsoft Certified: Azure Fundamentals May 2025 ([click](#))

.NET Development - Foundation Jan 2024 ([click](#))